
Neurorrhaphy of the Recurrent Laryngeal Nerve

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- BACKGROUND:** It has been controversial to anastomose the severed recurrent laryngeal nerve after transection because when severed recurrent nerves are directly anastomosed, the vocal cords do not regain normal movements and are fixed in the median. The objective of this study was to learn whether direct neurorrhaphy is necessary if the recurrent laryngeal nerve is severed during thyroid or parathyroid surgery.
- STUDY DESIGN:** From 1998 to 2001, 12 patients who had a complete recurrent laryngeal nerve injury during thyroid or parathyroid surgery were enrolled into this study. Eight had primary repair of the nerve; four did not have repair because of cancer invasion. Patients were followed with laryngoscopic or laryngovideostroboscopic examination at 3 months and 6 months postoperatively. Subjective ratings of aspiration and voice quality were based on patient reports. Perceptual voice quality was rated according to grade, roughness, breathiness, asthenia, and strain (GRBAS) scales. If a patient's voice quality, aspiration, and GRBAS scales did not improve at 6 months postoperatively, medialization laryngoplasty was considered.
- RESULTS:** Eight patients with immediate repair showed improved voice quality, aspiration, GRBAS scales, and maximum phonation time at 6 months after surgery. A significant decrease of glottal gap was also noted in patients with neurorrhaphy but not in four patients without neurorrhaphy. Medialization laryngoplasty was performed in all four patients without neurorrhaphy in the followup period. Immobilization of the vocal cord was noted in all of them, but only one in the neurorrhaphy group had an atrophy of the cord; all four patients without neurorrhaphy had atrophy.
- CONCLUSIONS:** Neurorrhaphy of the recurrent laryngeal nerve is a simple and effective procedure to improve voice quality, aspiration, GRBAS scales, glottal gap, and maximum phonation time by preventing atrophy of the vocal cord. (J Am Coll Surg 2003;197:52–57. © 2003 by the American College of Surgeons)
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Recurrent laryngeal nerve (RLN) palsy is a major complication after thyroid and parathyroid surgery, although in an experienced hand the rate of injury should be quite low: from 0.3% to 13.2%.^{1–3} Unilateral paralysis causes hoarseness, shortening of phonation, and aspiration. Direct anastomosis rarely results in recovery of movement of the vocal cord. Aberrant regeneration of adductor and abductor fibers accounts for this lack of function.^{4,5} We report results of neurorrhaphy in treating recurrent la-

ryngeal nerve injury that occurred during thyroid or parathyroid surgery and compare them with results in patients without neurorrhaphy.

METHODS

From January 1998 to December 2001, 12 patients who had a complete RLN injury or whose unilateral RLN had to be sacrificed because of various diseases were enrolled into this study. Eight had a primary repair of the RLN under an operating microscope. Anastomoses were usually made with two stitches of 10-0 nylon. Four of them did not have a nerve repair because of cancer involvement of the distal stump. An endocrine surgeon judged the completeness of the injury and a plastic surgeon did the neurorrhaphy.

Subjective ratings of aspiration and voice quality were based on patient reports. Ratings of aspiration were categorized as no aspiration, 0; mild, 1; moderate, 2; or

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Table 1. Patient Data

Pt. No.	Age (y)	Gender	Disease/procedure	Neurorrhaphy	Followup (mo)
1	40	F	Papillary cancer/total thyroidectomy	Yes	17
2	45	F	Recurrent nodular goiter/lobectomy	Yes	6
3	45	M	Secondary hyperparathyroidism/parathyroidectomy	Yes	6
4	19	F	Hyperthyroidism/subtotal thyroidectomy	Yes	9
5	47	F	Nodular goiter/lobectomy	Yes	10
6	52	M	Secondary hyperparathyroidism/parathyroidectomy	Yes	20
7	39	F	Secondary hyperparathyroidism/parathyroidectomy	Yes	6
8	52	F	Papillary cancer/total thyroidectomy	Yes	7
9	33	M	Papillary cancer/total thyroidectomy	No	6
10	48	F	Papillary cancer/total thyroidectomy	No	6
11	56	F	Papillary cancer/total thyroidectomy	No	6
12	75	F	Papillary cancer/total thyroidectomy	No	24

severe, 3. The mild rating was given to subjective reports of occasional choking requiring no dietary modifications. Aspiration was considered moderate when symptoms necessitated dietary modifications. The rating of severe aspiration was applied in circumstances in which feeding tube placement was required for alimentation.⁶ Ratings of voice quality were grouped into categories: markedly improved, 3; improved, 2; unchanged, 1; and worse, 0. A speech pathologist and senior laryngologist rated perceptual voice quality according to grade, roughness, breathiness, asthenia, and strain (GRBAS).^{7,8} Ratings were carried out during patients' sustained "i" and "a" phonations recorded by the laryngostroboscopic video. GRBAS scales were rated on a 5-point scale in which 0 was normal, 1 was mild dysphonia, 2 indicated moderate dysphonia, 3 was severe dysphonia, and 4 was aphonia. Each patient underwent laryngoscopy or laryngovideoscopy and recording 3 months and 6 months after the RLN injury or neurorrhaphy. Stroboscopy was performed with an Olympus ENF, Type L3 fiberoptic laryngoscope or a Karl Storz 70 rigid scope mated to a laryngostroboscope (model 8100; Kay Elemetrics Inc). The relative size of the glottal gap was assessed on a 5-point scale. Rating forms were as follows: no gap during phonation, 0; minimal gap, 1; small gap extending up to one-third of the posterior membranous vocal folds, 2; moderate gap extending up to two-thirds of the posterior membranous vocal folds, 3; and complete glottal incompetence with no observable contact between the vocal folds, 4.⁹ Atrophy of the vocal cord was re-

corded by one of the authors (laryngologist) after laryngoscopic or laryngovideoscopic examination. Maximum phonation time was measured with a sustained vowel (a) at comfortable pitch and loudness.

If patients' breathy and harsh phonation and life-threatening aspiration did not improve after 6 months, medialization laryngoplasty to correct glottal incompetence was performed.¹⁰ But one patient had medialization laryngoplasty 24 months after injury.

Statistical analysis

Statistical analyses were performed using the Wilcoxon signed-rank test for paired observations of ordinal variables, and Fisher's exact test for categorical (binary) data. A *p* value of less than 0.5 was considered significant. The SPSS package (SPSS Inc) was used for the statistical analysis.

RESULTS

Clinical data for 12 patients are listed in Table 1. There were nine women and three men included in this study. Ages ranged from 19 to 75 years (mean 45.9 ± 13.5 years [SD]) and followup period ranged from 6 to 24 months (mean 10.5 ± 6.4 months). Eight patients with neurorrhaphy had improvement of GRBAS scales at 6 months postoperatively, as compared with those at 3 months ($p_1 = 0.008$, $p_2 = 0.023$, $p_3 = 0.038$, $p_4 = 0.014$, and $p_5 = 0.034$, respectively) (Figs. 1–5). Four patients without neurorrhaphy had no significant improvement of GRBAS scales at 6 months postopera-

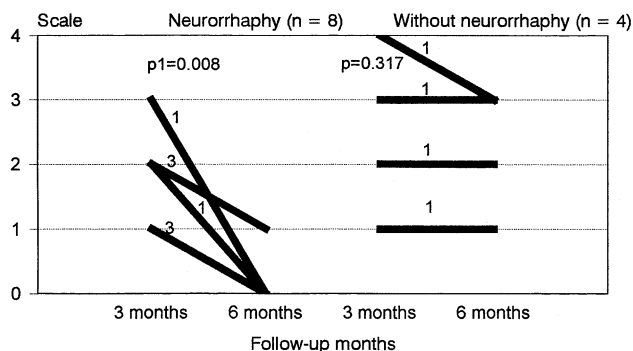


Figure 1. Perceptual rating of voice quality according to grade scales, between patients with (n = 8) and without (n = 4) neurorrhaphy on a 5-point scale in which 0 was normal; 1 was mild dysphonia; 2 was moderate dysphonia; 3 indicated severe dysphonia; and 4 was aphonia. Number shown with each line indicates the number of patients with the results depicted by the line.

tively, as compared with those at 3 months (Figs. 1–5). Subjective ratings of aspiration also improved in eight patients with neurorrhaphy at 6 months ($p = 0.008$), as compared with those at 3 months (Fig. 6), as did the subjective ratings of voice quality ($p = 0.010$) (Fig. 7). But subjective ratings of aspiration and voice quality did not change significantly in four patients without neurorrhaphy (Figs. 6 and 7).

The glottal gap decreased significantly at 6 months postoperatively in eight patients with neurorrhaphy as compared with that at 3 months ($p = 0.011$) (Fig. 8). The glottal gap did not change significantly at 6 months postoperatively in four patients without neurorrhaphy as compared with that at 3 months (Fig. 8). Medialization laryngoplasty was necessary in these four patients to

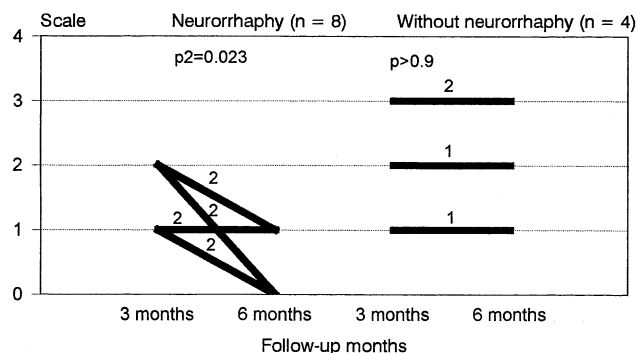


Figure 2. Perceptual rating of voice quality according to roughness scales between patients with (n = 8) and without (n = 4) neurorrhaphy on a 5-point scale in which 0 was normal; 1, mild roughness; 2, moderate roughness; 3, severe roughness; and 4, complete roughness. Number shown with each line indicates the number of patients with the results depicted by the line.

correct the glottal incompetence. Maximum phonation time increased from 4.9 ± 1.3 seconds to 10 ± 1.8 seconds in eight patients with neurorrhaphy ($p = 0.011$). But the maximum phonation did not change significantly in patients without neurorrhaphy. All 12 patients had immobilization of the vocal cord, but only 1 in 8 patients with neurorrhaphy had an atrophy of the vocal fold. All four patients without neurorrhaphy had atrophy of the vocal fold before medialization laryngoplasty was performed. The differences between the two groups were significant ($p = 0.010$).

DISCUSSION

Various factors that may increase the risk of injury to the RLN during thyroid and parathyroid surgery include

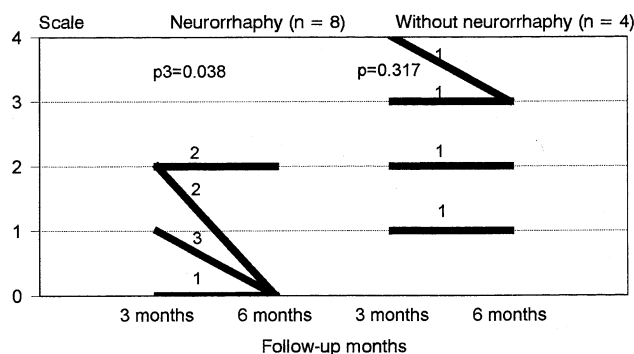


Figure 3. Perceptual rating of voice quality according to breathiness scales between patients with (n = 8) and without (n = 4) neurorrhaphy on a 5-point scale in which 0 was normal; 1, mild breathiness; 2, moderate breathiness; 3, severe breathiness; and 4, complete breathiness. Number shown with each line indicates the number of patients with the results depicted by the line.

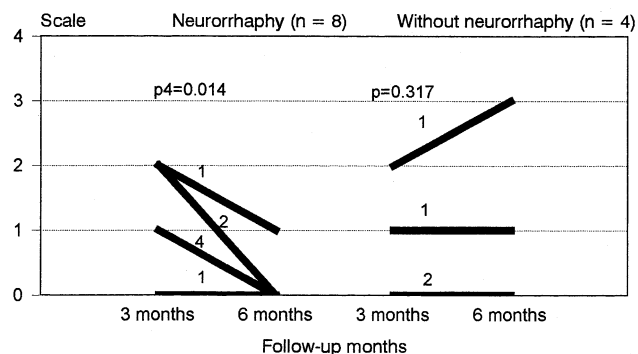


Figure 4. Perceptual rating of voice quality according to asthenia scales between patients with (n = 8) and without (n = 4) neurorrhaphy on a 5-point scale in which 0 was normal; 1, mild asthenia; 2, moderate asthenia; 3, severe asthenia; and 4, complete asthenia. Number shown with each line indicates the number of patients with the results depicted by the line.

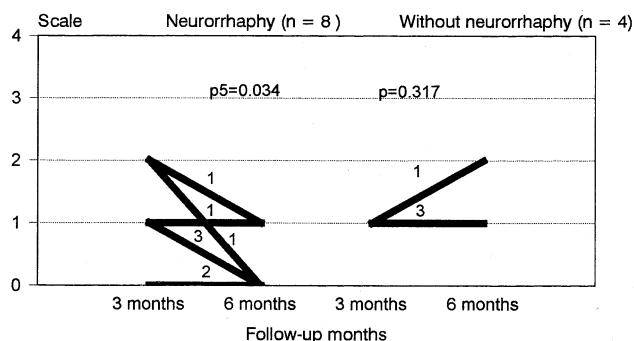


Figure 5. Perceptual rating of voice quality according to strain scales between patients with (n = 8) and without (n = 4) neurorrhaphy on a 5-point scale in which 0 was normal; 1, mild strain; 2, moderate strain; 3, severe strain; and 4, complete strain. Number shown with each line indicates the number of patients with the results depicted by the line.

variation in anatomy, extensive local and regional disease, previous surgery, and previous radiation.¹¹ In this study, the causes were total thyroidectomy for thyroid cancer (n = 6), repeated surgery (n = 1), by accident (n = 2), and parathyroid surgery (n = 3). Here, on Taiwan, alcohol injection of the parathyroid gland for secondary hyperparathyroidism was applied in several nephrology clinics. Patients finally had to have parathyroidectomy because of symptomatic secondary hyperparathyroidism. In that situation, severe adhesion was found between the RLN and parathyroid glands. Three RLN injuries were found during parathyroid surgery in this series.

The literature on laryngeal reinnervation is extensive. An excellent review of the subject was published by Rice.¹² But it has been considered controversial to re-

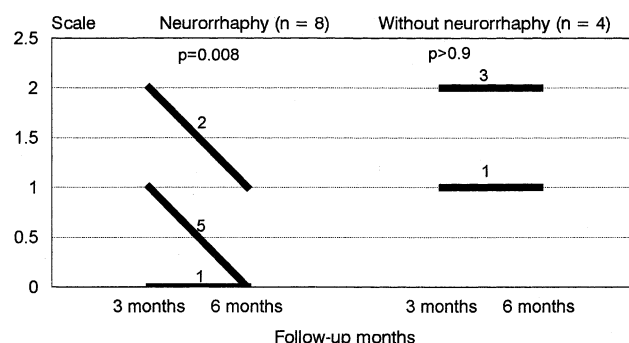


Figure 6. Subjective rating of aspiration according to patient reports between patients with (n = 8) and without (n = 4) neurorrhaphy on a 4-point scale in which 0 was normal; 1, mild; 2, moderate; and 3, severe aspiration. Number shown with each line indicates the number of patients with the results depicted by the line.

anastomose a severed RLN after transection.¹³⁻¹⁵ Frazier and Mosser¹⁶ anastomosed the recurrent laryngeal nerve to the ansa hypoglossi in 10 patients and found improvement of paralysis in 60% overall.

Hoover¹⁷ stated that attempts at restoration of recurrent laryngeal function by nerve anastomoses had failed. Siribodhi and colleagues⁴ confirmed that no functional movement was found despite return of electromyographic activity after repair of the laryngeal nerve, and they proposed that aberrant regeneration of adductor and abductor fibers accounted for the lack of normal function. But voices usually improved because the reinnervated cords recovered from atrophies and restored tension during phonation.¹⁸ Aspiration may also lessen after a direct anastomosis of the RLN.¹⁹ Boles and Fritzel¹³ studied injury and repair of the RLN in dogs and concluded that surgical repair of the RLN could possibly

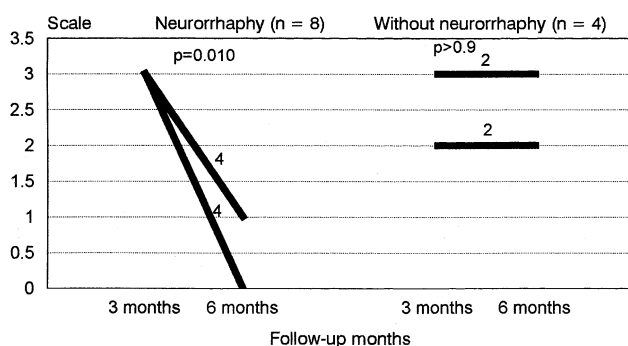


Figure 7. Subjective rating of voice quality improvement according to patient reports between patients with (n = 8) and without (n = 4) neurorrhaphy on a 4-point scale in which 0 was markedly improved; 1, improved; 2, unchanged; and 3, worse. Number shown with each line indicates the number of patients with the results depicted by the line.

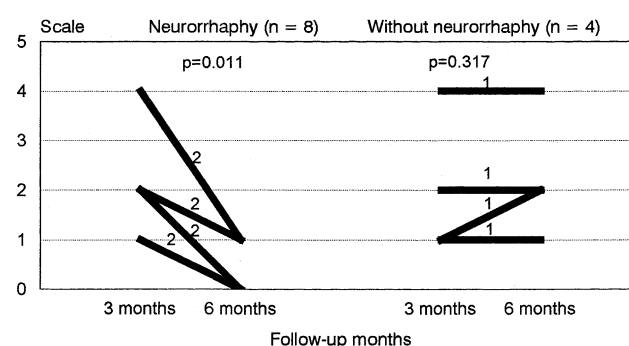


Figure 8. Laryngoscope or laryngovideoscopes rating of glottal gap between patients with (n = 8) and without (n = 4) neurorrhaphy on a 5-point scale, in which 0 was normal; 1, minimal; 2, moderate; 3, severe; and 4, complete glottal incompetence. Number shown with each line indicates the number of patients with the results depicted by the line.

return tone and some adduction of the cord and improve the glottis as a phonator and a protective sphincter.

In the literature, treatment of RLN paralysis after thyroidectomy has centered on postoperative treatment. This might be because of the fact that nerves may unknowingly be traumatized during operations and paralysis might not be discovered until much later in the postoperative period. This might give the mistaken impression that neurorrhaphy should not be done in case of nerve injury. Most earlier data seem to rule against RLN anastomosis despite a case report of the good recovery after primary anastomosis.¹¹ Despite what seems like a fair number of successful results, the consensus over the years has been to disregard neurorrhaphy.⁵ Some surgeons would make no attempt to repair the nerve, even if it was partially resected because of involvement with carcinoma of the thyroid. Some surgeons have advocated using the ansa-to-recurrent laryngeal nerve anastomosis to improve phonation.²⁰⁻²⁶ The ansa-to-RLN anastomosis should be considered if direct anastomosis causes too much tension. But this procedure takes 1 to 2 hours and needs some strap muscle dissection.²⁵ Routine exploration of the RLN is our policy for thyroid and parathyroid surgery in this department. The nerve injury was found during operation and neurorrhaphy was accomplished in eight patients. Local eradication of the gross tumor, even involving resecting the RLN, should be considered to ensure adequate resection. Four patients did not receive reanastomosis because of resection of papillary thyroid cancer at the entry of the RLN to the cricothyroid membrane. All four patients received ¹³¹I treatment and lifelong thyroid replacement therapy, and their thyroid stimulating hormone and thyroglobulin levels were suppressed in the followup period. If a nerve is injured during thyroid or parathyroid surgery, we use 2x loupes to identify both side of the RLN and to keep the nerve in its natural position. An anastomosis is then performed with two stitches of 10-0 nylon under an operating microscope.

All eight patients with direct neurorrhaphy had recoveries of voice quality, aspiration, GRBAS scales, and maximum phonation time. Although immobilization of the vocal cord was found in all patients with nerve injuries in the followup period of 6 to 24 months, only one patient had atrophy of the vocal cord in the neurorrhaphy group, and all patients in this group had improved glottal closure, increased vocal bulk, and improved voice quality. We think the diminished glottal

gap might be from lack of atrophy of the cord and compensation of the contralateral side. On the contrary, none of the four patients without neurorrhaphy recovered voice quality, aspiration, glottal gap, GRBAS scales, and maximum phonation time. All four patients had atrophy of the vocal cord and needed medialization laryngoplasty to close the glottal gap and to improve voice quality, aspiration, GRBAS scales, and maximum phonation time.¹⁰

In conclusion, direct neurorrhaphy is a simple and effective procedure to improve voice quality, aspiration, GRBAS scales, and maximum phonation time in the followup period. Neurorrhaphy is necessary if the anastomosis can be done easily in case of nerve injury to prevent atrophy of the vocal cord. Ansa-to-RLN anastomosis or grafting should be considered if the defect extends too far and if too much tension is found after direct neurorrhaphy. To ensure adequate resection, local eradication to remove the thyroid cancer grossly and even to sacrifice the nerve should be considered. But medialization laryngoplasty may be necessary later after ¹³¹I treatment.

Author Contributions

Study conception and design: Chou

Acquisition of data: Lu, Hsu

Analysis and interpretation of data: Chou

Drafting of manuscript: Chou

Critical revision: Su, Jeng

Statistical expertise: Chou

Obtaining funding: Chou

Supervision: Su

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